



Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

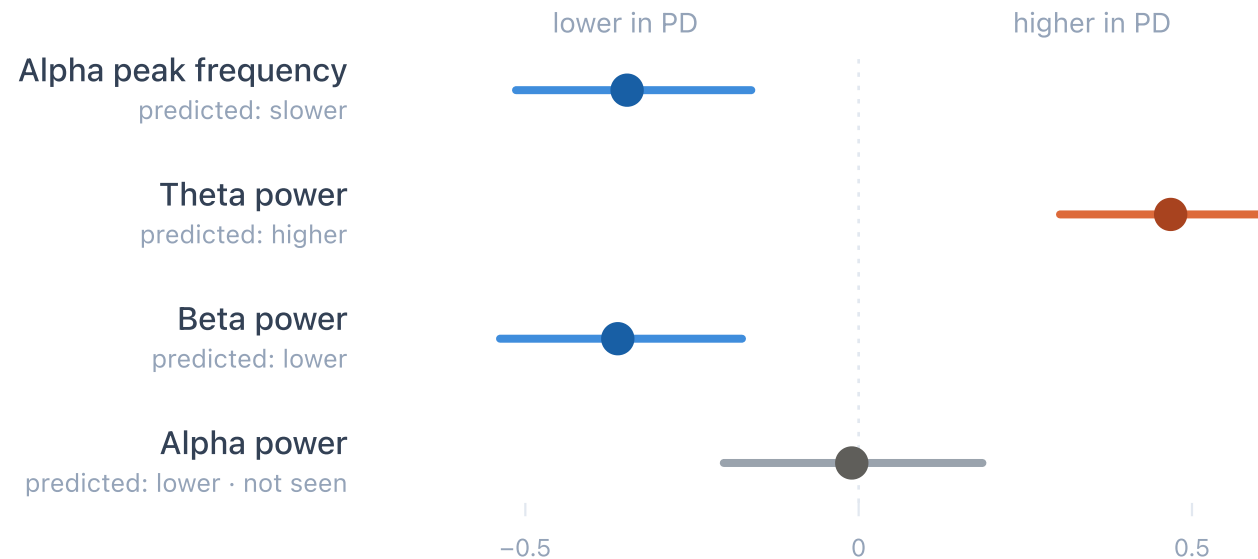
Hypotheses

Compared to age-matched healthy controls, Parkinson's patients at rest show:

- **slowing of the alpha peak frequency** (8 to 13 Hz)
- **increased theta-band power** (4 to 8 Hz)
- **decreased cortical beta-band power** (13 to 30 Hz)

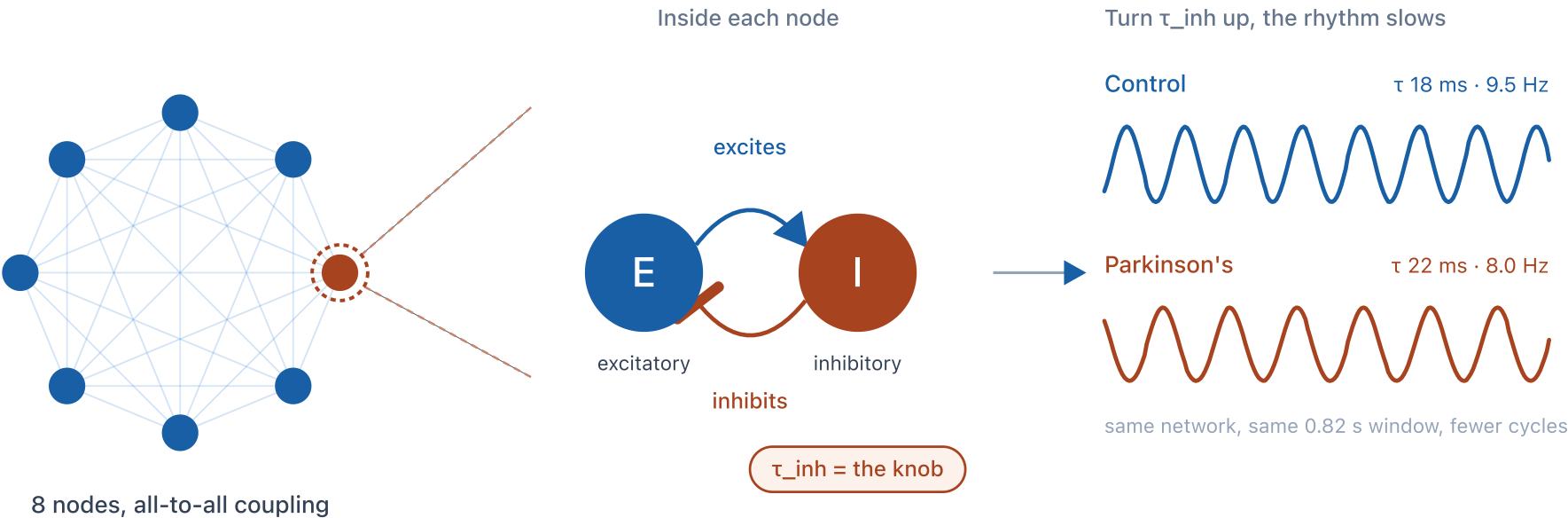
How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval



The model: a Wilson-Cowan network

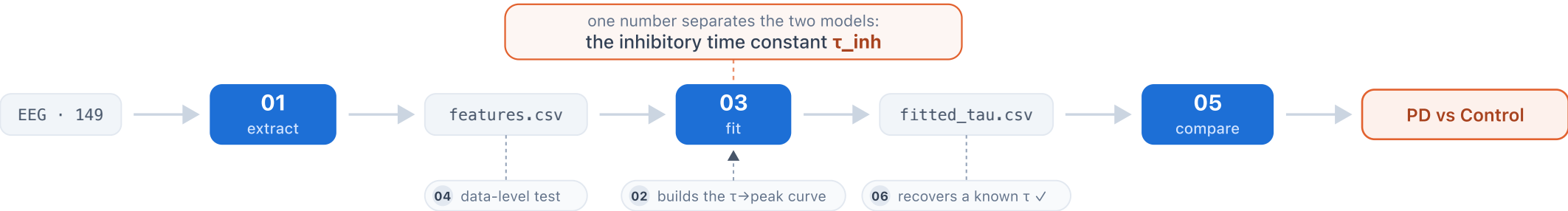
8 coupled cortical nodes, each an excitatory-inhibitory loop. τ_{inh} is the only knob we turn.



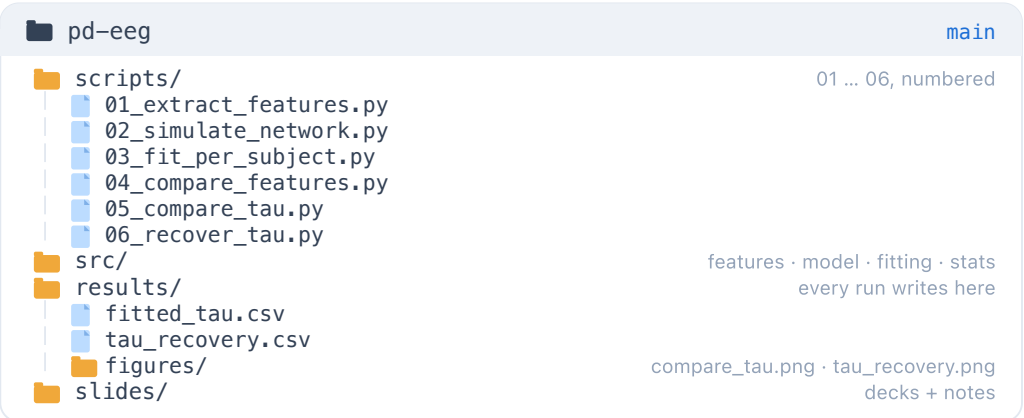
The final adaptation from healthy to patient is a single parameter: a longer inhibitory time constant τ_{inh} , about **+3.25 ms** in PD.

Code & repository

THE PIPELINE EEG to result, through six numbered scripts



THE REPOSITORY

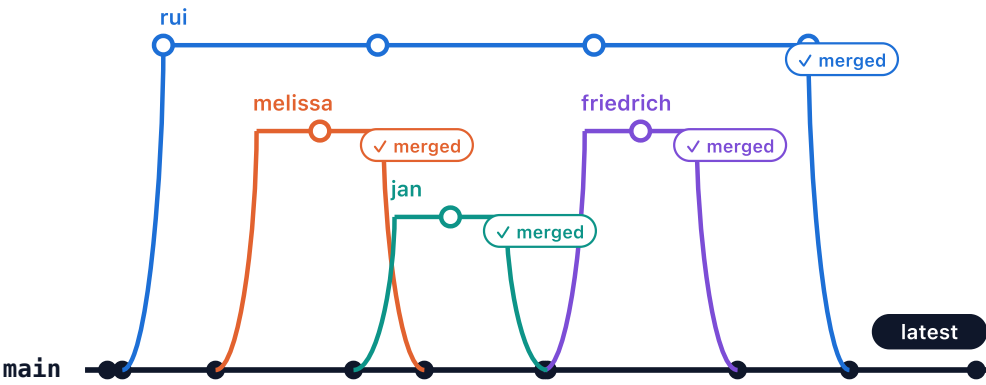


every result is one script run

```
bash
$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.00025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias -0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

VERSION CONTROL

feature branches, reviewed, merged into main



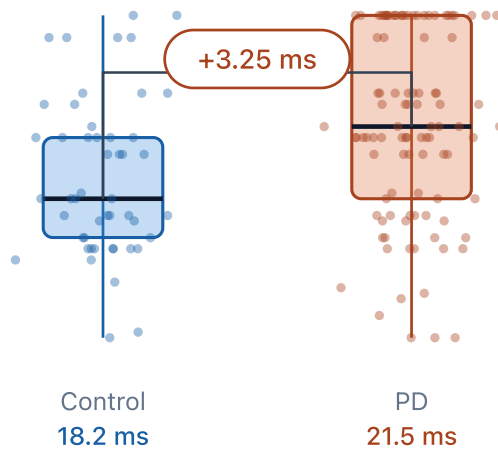
50 commits · feature branches → reviewed → merged into main · fully reproducible

Does the model separate patients from controls?

Fitted inhibitory time constant

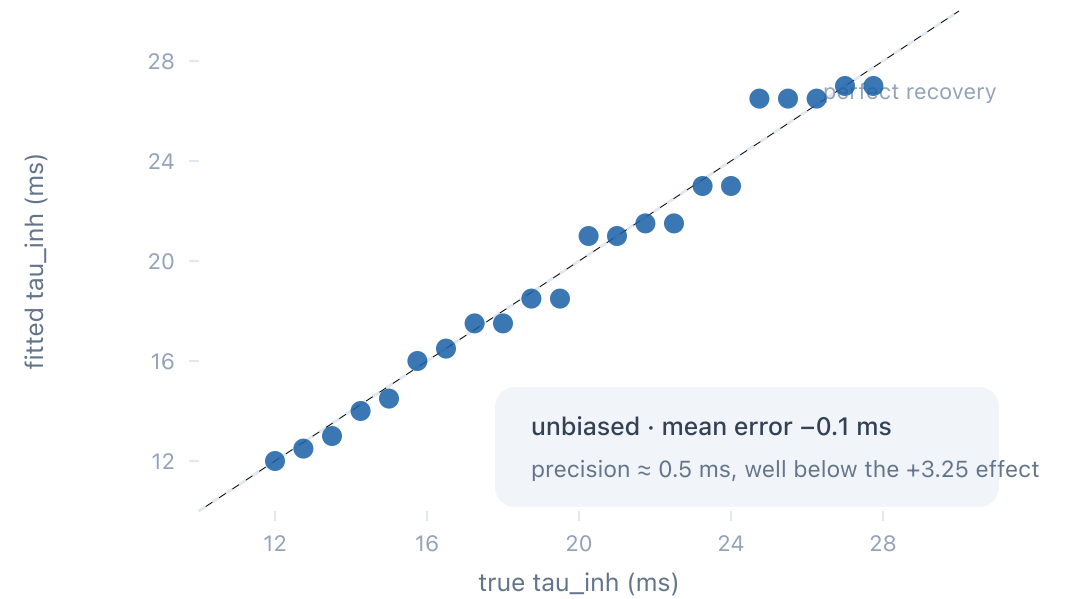
the model parameter we change for a patient

fitting recovers tau to ± 0.5 ms, far below this group gap



Validation: recovering a known parameter

simulate at a known tau, fit it back



Fitted τ_{inh} is **+3.25 ms** higher in PD (one-sided Mann-Whitney, $p \approx 0.0002$). The recovery test shows this is a real separation, not an artefact of the fitting.

Outlook and open challenges

CURRENT CHALLENGES

- One parameter (τ_{inh}) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted τ to clinical scores and to known inhibitory (GABAergic) changes in PD

What we took away

Rui [your biggest learning]

Jan [your biggest learning]

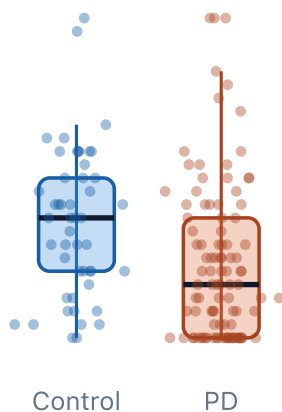
Melissa [your biggest learning]

Friedrich [your biggest learning]

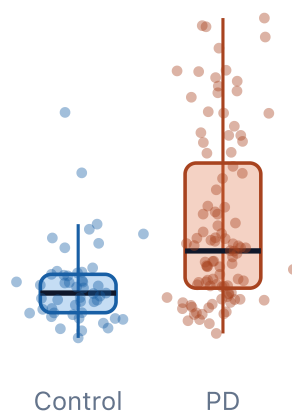
Appendix: per-subject distributions

PD vs Control: per-subject EEG features

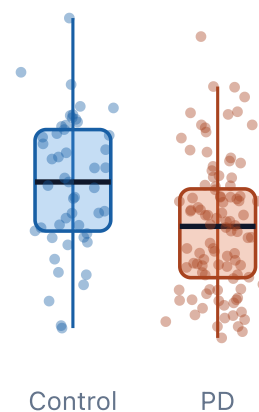
Alpha peak frequency
slower in PD



Theta power
higher in PD



Beta power
lower in PD



Alpha power
unchanged

